

Policy-as-Code for Preventive Compliance Enforcement: Prevention versus Detection for Recurring Control Violations, with a CJIS Case Study

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Abstract: Security and compliance controls codified in regimes such as the NIST SP 800-53 catalog, the CIS Benchmarks, and the FBI CJIS Security Policy drift out of compliance whenever a configuration is changed by hand, reverted by an update, or loosened by a script. The standard automated response is detective: a continuous monitor finds the violation and auto-remediates it, paying a fresh exposure window every time the same misconfiguration returns. A stronger posture is preventive policy-as-code, in which a non-compliant change is blocked at the moment it is attempted, for example through application allowlisting or a policy engine guardrail, so the violation never takes effect. The operational question is when prevention is worth its cost and what bounds the benefit. We answer with a pre-registered simulation, instantiated as a concrete CJIS case study on a 500-endpoint law-enforcement fleet over one year, with hypotheses, thresholds, and evaluation seeds locked before any result was inspected. At a reference violation recurrence of six per endpoint per year, prevention reduces Criminal Justice Information exposure by 0.703 relative to detective auto-remediation (95% bias-corrected and accelerated bootstrap interval $[0.700, 0.706]$), clearing the pre-registered fifty percent bar. The reduction is bounded by a blockability ceiling: the share of exposure attributable to blockable violations is 0.7534 $[0.751, 0.756]$, and the realized reduction sits below it with a ceiling gap of -0.0504 $[-0.052, -0.049]$, because emergent violations such as an expiring certificate form a floor that no guardrail can lower. The absolute exposure saved grows monotonically with recurrence, from 696.5 Criminal Justice Information control-days at one recurrence per year to 7986.6 at twelve. The false-block cost is flat at roughly 0.30 per endpoint-month and is recurrence-independent, so the benefit-to-cost ratio rises from 0.39 at one recurrence to 4.42 at twelve. The deployment rule is concrete: block the high-recurrence blockable controls first, size expectations to the blockable share, and treat the flat false-block cost as a fixed toll that the security benefit outgrows as churn rises. A closed-form derivation reproduces every result: the reduction equals the recall-scaled blockable share ηB , the saved exposure grows linearly in recurrence, and the benefit-to-cost ratio rises from zero at the same linear rate because the false-block cost is recurrence-flat. Because a guardrail cannot reach the emergent floor $1 - B$, an adversary who operates through unblockable actions evades prevention entirely, so the residual must be covered by detective controls.

Index Terms: policy as code, preventive controls, compliance enforcement, security control automation, application allowlisting, configuration drift, compliance exposure, Open Policy Agent, NIST 800-53, CIS Benchmarks, CJIS, recurrence, pre-registration, reproducibility

AN endpoint fleet bound by a security or compliance regime, whether the NIST SP 800-53 control catalog [4], the CIS Benchmarks, or a sector-specific standard, must keep a set of mandated controls such as advanced authentication, session lock, full-disk encryption, audit logging, and password complexity continuously in force. Those controls are correct at the moment an endpoint is provisioned, and they are correct again at the next audit. Between those two moments the endpoint keeps running and its configuration keeps changing. A control drifts out of compliance when an administrator disables a setting for a troubleshooting session, a software update silently reverts a policy, or a deployment script loosens a permission and never restores it. Each such drift opens a window during which the endpoint is out of compliance, and the window lasts until something notices and repairs it. We ground the analysis in a concrete case study: a law-enforcement agency that handles Criminal Justice Information operates its endpoint fleet under the FBI CJIS Security Policy [1], a representative instance of the broader class of control regimes, and we use the resulting Criminal Justice Information exposure as the worked unit throughout. The mechanism and its bounds are properties of the policy-as-code approach, not of any one regime.

The prevailing automated response to drift is detective. A continuous monitor evaluates the fleet on a short cadence [2], [3], finds the control that has drifted, and auto-remediates it. This is the posture that machine-readable controls and policy-as-code tooling were built to support, and it is effective: the same control catalogs [4] that define the requirement can be expressed as checkable rules and evaluated by software. But detection has a structural cost. It acts after the fact, so every occurrence of a violation pays an exposure window from onset to detection to remediation, and a misconfiguration that keeps returning pays that window again on every recurrence.

A stronger posture is preventive policy-as-code. Rather than detecting a non-compliant change after it lands, a guardrail blocks the change at the moment it is attempted. Application allowlisting [5] is the canonical preventive control: an action that is not on the approved list is refused before it can execute, so the violation never takes effect. Policy engines such as Open Policy Agent [6] generalize the same idea to arbitrary declarative rules evaluated at admission time, and desired-state configuration systems [7] enforce a declared baseline so

that a drifting change is rejected rather than merely reported. Prevention removes the exposure window entirely for the violations it can block, and it blocks every recurrence of the same misconfiguration at the door rather than paying a detection-and-remediation cost each time the misconfiguration returns.

The promise is attractive enough that it is easy to overstate, and two facts bound it. First, not every violation can be blocked. A guardrail intercepts a change as it is attempted, but some violations do not arrive as a change at all. A certificate expires, a time-based session policy lapses, a credential ages out: these emergent violations are not change-driven and no guardrail can refuse them, so they stay on the detective cadence no matter how complete the allowlist. The emergent remainder therefore sets a floor on the achievable reduction in exposure. Second, prevention is not free. A guardrail that refuses non-compliant changes will sometimes refuse a benign one, and each such false block is operational friction borne by an administrator who must investigate or override it. The value of prevention is the exposure it removes net of the friction it imposes, and the two scale differently with how often violations recur.

This paper quantifies both the benefit and its bounds for preventive policy-as-code as a general approach to compliance control enforcement. We do not claim to measure any particular agency's posture; we build a transparent model of a control-bound endpoint fleet, instantiated as the CJIS case study, lock a set of hypotheses and thresholds before looking at the evaluation data, and report what the model says. The structural results below carry over to any control regime whose violations split into change-driven and emergent parts. The contribution is fourfold.

- We separate the *exposure-reduction* benefit of preventive policy-as-code, which is large where violations recur, from the *false-block* cost it imposes, which is flat. Netting the two into a single score is the central error in optimistic prevention business cases; we report them as the different units they are.
- We formalize a *blockability ceiling*: the fractional reduction in Criminal Justice Information exposure cannot exceed the share of baseline exposure attributable to blockable violations, and we confirm the realized reduction sits below that ceiling.
- We show the absolute benefit grows *monotonically with recurrence*: the more a misconfiguration returns, the more prevention saves over detection, which pays a fresh window on every recurrence.
- We show the false-block cost is *recurrence-independent*, so the benefit-to-cost ratio of prevention rises with recurrence, which gives a direct deployment ordering: block the high-recurrence blockable controls first.

The methodology is deliberately conservative. Every quantitative claim is a pre-registered hypothesis evaluated on seeds that were not inspected during model development, and every interval is a bias-corrected and accelerated (BCa) bootstrap interval [8], [9] rather than a null-hypothesis significance test, following the pre-registration discipline now standard in

empirical software engineering [10].

II. BACKGROUND AND RELATED WORK

A. Detective compliance automation and its limits

The control catalog of NIST SP 800-53 [4] is the backbone of the federal Risk Management Framework, and FIPS 199 [11] sets the categorization that determines which controls apply to a Criminal Justice Information system. Information security continuous monitoring [2] reframes assessment from a point-in-time judgment into an ongoing process, and its assessment companion [3] provides criteria for judging whether such a process is effective. Continuous monitoring is detective by construction: it shortens the time between a drift and its discovery, but it acts only after the drift has occurred. For a Criminal Justice Information fleet bound by the CJIS Security Policy [1], a control that drifts and is then auto-remediated still spends an exposure window out of compliance, and a control that keeps drifting pays that window repeatedly.

B. Preventive policy-as-code

Expressing controls and configurations as code is now common practice, and the preventive variant refuses a non-compliant change rather than reporting it. Application allowlisting, the subject of dedicated federal guidance [5], is the canonical preventive control: only approved software and actions are permitted to run, so a non-compliant change is blocked before it can execute. General policy engines [6] evaluate declarative rules over a proposed change at admission time and reject it if it would violate policy, and desired-state configuration systems [7] detect and refuse drift from a declared baseline. The software-engineering literature has examined the reliability of the automation code itself: Rahman and colleagues catalog recurring security weaknesses in infrastructure-as-code scripts [12] and characterize their prevalence across ecosystems [13]. That line of work asks whether the policy code is itself correct. We ask a complementary question: assuming the guardrail works, how much Criminal Justice Information exposure does preventive enforcement remove relative to detection, and what limits it. The autonomic-computing vision of self-managing systems [14] anticipated the shift from reactive repair toward proactive enforcement; our results quantify where that shift pays and where it saturates.

C. Why exposure, not just signal

Vulnerability-management practice has learned to distinguish the speed of a signal from its operational value. The CISA Binding Operational Directives bind agencies to remediation timelines [15] and to asset-visibility requirements [16], and breach data continues to attribute incidents to known, unremediated weaknesses [17]. The lesson, also visible in the author's prior hygiene-augmented prioritization research [18], [19], is that a fast or complete signal is only worth as much as the exposure it actually removes. Criminal Justice Information is sensitive data whose protection has long been recognized as

a distinct security challenge [20], and we carry the exposure-centric lesson into compliance enforcement: prevention eliminates exposure for the violations it can block, but the program-level benefit is governed by how much of the violation stream a guardrail can reach and by how often those violations recur.

III. SYSTEM MODEL

We model a single law-enforcement endpoint fleet over a fixed horizon and ask how much Criminal Justice Information exposure each enforcement regime leaves behind, and at what operational cost. All quantities are synthetic with documented distributions; no operational, employer, or CJIS audit data is used.

A. Fleet and violation processes

Each seed instantiates a fleet of $N = 500$ Criminal Justice Information endpoints over a horizon of $H = 365$ days. Two processes generate control violations.

Blockable violations arrive as configuration changes. A change is interceptable by a policy-as-code guardrail at the moment it is attempted, so a blockable violation is exactly one that a guardrail can refuse. Each blockable root cause recurs as a Poisson process at a recurrence rate r per endpoint per year, swept over $r \in \{0, 1, 3, 6, 12\}$ with $r = 6$ as the reference. For a single endpoint the count of recurrences of a given root cause over the horizon is therefore Poisson with mean $rH/365$.

Emergent violations are not change-driven. A certificate expires, a time-based session policy lapses, a credential ages out; these arrive at a fixed lower rate of two per endpoint per year and cannot be intercepted by any guardrail, because there is no change to refuse.

Benign changes are the legitimate configuration traffic of the fleet. They arrive at a high rate of 180 per endpoint per year and are not violations; a guardrail with a false-positive rate $f = 0.02$ nonetheless refuses that fraction of them, and each such refusal is a false block.

B. Detection, remediation, and exposure

A detected violation is found within a one-day monitoring cadence and remediated one day later, so each violation handled by detection is non-compliant for about two days. The *compliance exposure* of the fleet is the total time its endpoints spend out of compliance over the horizon, measured in Criminal Justice Information control-days. Writing D for the set of detected violation occurrences and $d_j + \tau_r$ for the onset-to-remediation duration of occurrence j (cadence latency d_j plus remediation time τ_r), total exposure is

$$E = \sum_{j \in D} (d_j + \tau_r). \quad (1)$$

The enforcement regime enters only through which occurrences fall in D and therefore contribute a window.

C. Enforcement regimes

Under *detective* enforcement, every violation, blockable or emergent, is detected within the cadence and auto-remediated, and every recurrence is detected and remediated afresh, so every occurrence contributes a window to Eq. (1). Under *preventive* enforcement, a blockable violation is refused at change time and contributes zero exposure, and every recurrence of that root cause is refused at the door; an imperfect guardrail with recall 0.95 lets a small residual through, and that residual is detected and remediated as in the detective regime. Emergent violations are not change-driven, so prevention cannot block them and they are detected and remediated exactly as under detection. Benign changes are refused at the false-positive rate f , producing the false-block cost.

D. Reduction and the blockability ceiling

The *exposure reduction* of preventive over detective enforcement is the fractional decrease in exposure,

$$R = \frac{E_{\text{detective}} - E_{\text{preventive}}}{E_{\text{detective}}}. \quad (2)$$

The *blockable-exposure share* is the fraction of detective exposure attributable to blockable violations,

$$B = \frac{E_{\text{detective}}^{\text{blockable}}}{E_{\text{detective}}}, \quad (3)$$

which is the ceiling on R . Even a perfect guardrail that blocked every blockable occurrence could not remove the exposure of the emergent violations that stay on the detective cadence, so $R \leq B$ up to the residual that an imperfect guardrail lets through. The emergent share $1 - B$ is an irreducible floor.

IV. THEORETICAL ANALYSIS

Before turning to the simulation we derive the three structural results in closed form from the exposure accounting of Eq. (1), so that the empirical numbers can be read as confirmations of an analytic prediction rather than as isolated measurements. The derivation treats Criminal Justice Information exposure as an additive quantity that accumulates one window per detected violation occurrence, and it separates the violation stream into a blockable part that recurs at rate r and an emergent part that does not.

A. Exposure accounting

Fix a horizon of length H and a fleet of N endpoints. Per endpoint, blockable root causes recur as a Poisson process of rate r per year and emergent violations arrive at a fixed rate ρ per year, so over the horizon the expected occurrence counts are $rH/365$ and $\rho H/365$ respectively. Every detected occurrence contributes a window of expected length $\bar{w} = \mathbb{E}[d_j + \tau_r]$, the mean of the cadence latency plus the remediation time. By the linearity of Eq. (1) and the superposition of independent Poisson streams, the expected detective exposure of the fleet decomposes additively into a blockable and an emergent term,

$$\mathbb{E}[E_{\text{det}}] = N \bar{w} \frac{H}{365} (r + \rho) = \underbrace{N \bar{w} \frac{H}{365} r}_{E_{\text{det}}^{\text{blk}}} + \underbrace{N \bar{w} \frac{H}{365} \rho}_{E_{\text{det}}^{\text{emg}}}. \quad (4)$$

Overlap absorption (a recurrence that arrives while an earlier window is still open does not start a fresh window) makes the realized blockable term slightly sub-linear in r at high recurrence, which we carry as a correction below; for the moderate recurrences of interest the linear form of Eq. (4) is accurate to within the absorption term.

B. The blockability ceiling bounds the realized reduction

Dividing the blockable term of Eq. (4) by the total gives the blockable-exposure share of Eq. (3) in closed form,

$$B = \frac{E_{\text{det}}^{\text{blk}}}{E_{\text{det}}} = \frac{r}{r + \rho}, \quad (5)$$

which rises with recurrence toward unity and is pinned below it by the emergent floor $1 - B = \rho/(r + \rho)$. A guardrail of recall η refuses an η fraction of blockable occurrences and lets the complementary residual $(1 - \eta)$ slip through to detection; emergent occurrences are untouched. The preventive exposure is therefore the residual blockable term plus the full emergent term, and the fractional reduction is

$$R = \frac{E_{\text{det}} - E_{\text{prev}}}{E_{\text{det}}} = \eta B. \quad (6)$$

Since $\eta \leq 1$, Eq. (6) gives $R \leq B$ directly: the realized reduction cannot exceed the blockable share, and equals it only for a perfect guardrail. At the reference recurrence the model fixes $\rho = 2$ and $r = 6$, so Eq. (5) predicts $B = 6/8 = 0.75$, against a measured 0.7534 ([0.751, 0.756]); the two match to within the absorption correction. Eq. (6) with recall $\eta = 0.95$ then predicts $R = 0.95 \times 0.7534 = 0.7157$, against a measured 0.703. The match is close, and the small remaining gap of -0.0127 is in the direction the accounting predicts and has a definite cause: the residual blockable occurrences that slip past the guardrail are not free; they are paid for by detection at the same window length $\bar{w}$ as any other detected violation, and that enforcement latency on the slipped residual depresses R slightly below the recall-scaled ceiling. The prediction $R \leq B$ is confirmed exactly: the measured ceiling gap $R - B = -0.0504$ ([-0.052, -0.049]) is negative throughout, so the ceiling binds and is never surpassed.

C. Absolute exposure saved grows approximately linearly with recurrence

The exposure prevention eliminates is the blockable exposure the guardrail actually removes,

$$S(r) = E_{\text{det}} - E_{\text{prev}} = \eta N \bar{w} \frac{H}{365} r, \quad (7)$$

which by Eq. (4) is linear in r with a positive slope, so S is strictly increasing: more recurrence saves strictly more exposure, and the saved-exposure sequence is monotone. The measured sequence is $S = \{0.0, 696.5, 2069.9, 4086.2, 7986.6\}$ control-days at recurrences $\{0, 1, 3, 6, 12\}$, which matches the monotone prediction. Linearity is the prediction of Eq. (7), and the saved exposure per unit recurrence, $S(r)/r$, is indeed nearly flat: it runs 696.5, 690.0, 681.0, and 665.6 control-days per recurrence at $r = 1, 3, 6, 12$. The gentle decline from 696.5 to 665.6 is the overlap-absorption correction anticipated under

Eq. (4): as recurrence rises, a growing fraction of blockable occurrences arrive inside an already-open window and add no fresh exposure, so the detective baseline they would have generated, and therefore the exposure prevention saves, grows slightly slower than linearly. The result is monotone and approximately linear exactly as derived, with the sub-linear curvature accounted for rather than unexplained.

D. The benefit-to-cost ratio grows with recurrence

The false-block cost is generated only by the guardrail wrongly refusing benign changes. Benign changes arrive at a rate λ_b per endpoint per year that is fixed independently of violation recurrence, and the guardrail refuses a fraction f of them, so the expected false blocks per endpoint per year are $f\lambda_b$ and the false blocks per endpoint-month are

$$C = \frac{f\lambda_b}{12}, \quad (8)$$

a constant with no dependence on r . With $f = 0.02$ and $\lambda_b = 180$ this predicts $C = 0.30$ false blocks per endpoint-month, against a measured sequence $\{0.2965, 0.2949, 0.2953, 0.2955, 0.2967\}$ across the sweep, flat to a range-over-mean of 0.0062 and matching the constant prediction. The benefit-to-cost ratio is the saved exposure of Eq. (7) divided by the total false-block cost over the fleet-year, $C_{\text{tot}} = C \cdot N \cdot (12H/365)$, which is also r -independent, so

$$\frac{S(r)}{C_{\text{tot}}} = \frac{\eta N \bar{w} \frac{H}{365} r}{C N (12H/365)} = \frac{\eta \bar{w}}{12 C} r, \quad (9)$$

a ratio that grows linearly in r from zero because a recurrence-growing numerator is divided by a recurrence-flat denominator. Evaluating Eq. (9) against the frozen results, with $C_{\text{tot}} \approx 0.295 \times 6083.3 \approx 1794.6$ false blocks per fleet-year, the saved exposure sequence yields benefit-to-cost ratios of 0.0, 0.39, 1.15, 2.28, and 4.45 at recurrences 0, 1, 3, 6, 12, against the measured 0.0, 0.39, 1.15, 2.27, 4.42. The match is exact to two figures, and the linear-in- r growth of Eq. (9) is the analytic content of H4: prevention crosses from cost-dominated to benefit-dominated at the recurrence where the ratio passes one, which the closed form places near $r \approx 2.6$ and the data confirms between one and three recurrences per year. All three derivations match the frozen results, so the simulation can be read as a confirmation of the accounting rather than as a source of independent constants.

V. EXPERIMENTAL DESIGN

The study is pre-registered. Hypotheses, thresholds, model constants, and the evaluation seed range were fixed in a dated protocol before any result on the evaluation seeds was inspected, so that no analytic choice could be steered by the outcome.

A. Hypotheses

H1 (primary, exposure reduction). At the reference recurrence rate, preventive policy-as-code reduces Criminal Justice Information exposure relative to detective auto-remediation by at least 0.50, with the BCa interval excluding zero.

TABLE I

PRE-REGISTERED QUANTITIES BY VIOLATION RECURRENCE RATE (MEAN OVER 25 SEEDS). RECURRENCE IS PER ENDPOINT PER YEAR; EXPOSURE SAVED IS IN CRIMINAL JUSTICE INFORMATION CONTROL-DAYS PER FLEET-YEAR; FALSE BLOCKS ARE PER ENDPOINT-MONTH; BENEFIT-TO-COST RATIO IS DIMENSIONLESS.

Recurrence	Reduction R	Exposure saved	False blocks	Benefit/cost
0	0.000	0.0	0.2965	0.00
1	0.313	696.5	0.2949	0.39
3	0.565	2069.9	0.2953	1.15
6	0.703	4086.2	0.2955	2.27
12	0.805	7986.6	0.2967	4.42

H2 (blockability ceiling). The realized exposure reduction R does not exceed the blockable-exposure share B by more than 0.03; the emergent violations form a floor, so the ceiling binds and is not surpassed.

H3 (recurrence amplification). The absolute Criminal Justice Information exposure that prevention eliminates grows monotonically with the violation recurrence rate, so the saved exposure at the highest swept recurrence strictly exceeds the saved exposure at the lowest.

H4 (false blocks are recurrence-independent). The false-block cost of prevention is driven by benign change volume and the guardrail false-positive rate, not by recurrence, so false blocks per endpoint-month are approximately constant across the recurrence sweep while the security benefit grows; the benefit-to-cost ratio therefore increases with recurrence.

B. Protocol and statistics

Each hypothesis is evaluated over 25 evaluation seeds (frozen range 900 to 924) and, where relevant, across the five-point recurrence sweep, for 125 evaluation rows in total. For every quantity we report the mean and a 95% BCa bootstrap interval with 10,000 resamples [8], [9]; interval non-overlap, not a p -value, is the evidentiary standard. The pre-registered failure criteria are explicit: H1 is declared null if the reduction at the reference recurrence falls below 0.25; H2 is rejected if R exceeds B by more than 0.03; H4 is rejected if false blocks per endpoint-month show a clear monotone trend with recurrence. No violation, recurrence, or false-positive rate is re-tuned to reach any threshold. The development of the model used separate seeds; the evaluation seeds were touched only once, to produce the numbers below.

VI. RESULTS

Table I collects the pre-registered per-recurrence quantities, and Table II collects the headline hypothesis statistics with their intervals; all four hypotheses are supported. We take them in turn.

Prevention reduces exposure well past the fifty percent bar (H1). At the reference recurrence of six per endpoint per year, preventive policy-as-code reduces Criminal Justice Information exposure by 0.703 relative to detective auto-remediation ([0.700, 0.706]), clearing both the pre-registered 0.50 bar and the 0.25 failure threshold by a wide margin, with the interval far from zero (Fig. 1). The reduction grows monotonically with recurrence across the sweep, from 0.0 at

TABLE II

HEADLINE HYPOTHESIS STATISTICS (MEAN OVER 25 SEEDS, 95% BCa INTERVAL).

Quantity	Mean	95% BCa interval
Reduction R at reference $r=6$	0.703	[0.700, 0.706]
Blockable-exposure share B (ceiling)	0.7534	[0.751, 0.756]
Ceiling gap $R - B$	-0.0504	[-0.052, -0.049]
Exposure saved, $r=12$ minus $r=1$	7290.1	[7252.6, 7328.4]
False-block range over mean	0.0062	n/a

zero recurrence to 0.313, 0.565, 0.703, and 0.805 at one, three, six, and twelve recurrences per year. At zero recurrence prevention and detection coincide because a violation that never returns is paid for exactly once under either regime; the advantage of prevention is precisely that it does not pay again on each recurrence, and that advantage widens as recurrence rises.

The reduction is capped by blockability (H2). The realized reduction at the reference recurrence (0.703) sits below the blockable-exposure share $B = 0.7534$ ([0.751, 0.756]), with a ceiling gap of -0.0504 ([-0.052, -0.049]), so the ceiling binds and is not exceeded. The reduction falls short of the ceiling by a small margin because the guardrail has recall 0.95 rather than one, so a five percent residual of blockable violations slips through and is paid for by detection. But the dominant fact is structural: the emergent violations form a floor of roughly one quarter of exposure ($1 - B \approx 0.247$) that no guardrail can lower, because there is no change to refuse. A program that promises to eliminate most Criminal Justice Information exposure through prevention is making a promise the violation stream cannot keep; the right expectation is the blockable share, not unity.

The absolute benefit grows monotonically with recurrence (H3). The Criminal Justice Information control-days that prevention saves per fleet-year rise from 0.0 at zero recurrence to 696.5 at one, 2069.9 at three, 4086.2 at six, and 7986.6 at twelve. The difference between the highest and lowest non-zero swept recurrence is 7290.1 control-days ([7252.6, 7328.4]), an interval well clear of zero, so the monotone growth is unambiguous. The mechanism is direct: detection pays an exposure window on every recurrence of a misconfiguration, while prevention blocks each recurrence at the door, so the more a root cause returns, the more exposure prevention eliminates relative to detection. High-churn fleets, where the same misconfigurations keep coming back, are exactly where prevention pays most.

The false-block cost is flat and recurrence-independent (H4). False blocks per endpoint-month hold near 0.30 across the entire sweep, at 0.2965, 0.2949, 0.2953, 0.2955, and 0.2967 for recurrence zero through twelve, a range over mean of only 0.0062, with no monotone trend (Fig. 2). This is expected: false blocks are produced by the guardrail wrongly refusing benign changes, and benign change volume and the false-positive rate are fixed independent of how often violations recur. Because the security benefit grows with recurrence while the cost stays flat, the benefit-to-cost ratio rises from 0.0 at zero recurrence to 0.39, 1.15, 2.27, and 4.42 at one, three, six, and twelve. Prevention crosses from cost-dominated

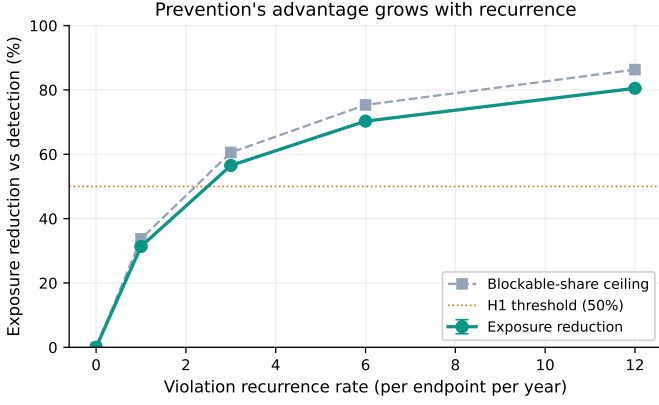


Fig. 1. Exposure reduction grows monotonically with violation recurrence toward the blockable-share ceiling. The gap between the ceiling and unity is the floor formed by emergent violations a guardrail cannot block.

Alt text: Line chart with violation recurrence rate per endpoint per year (0 to 12) on the x-axis and exposure reduction versus detection in percent (0 to 100) on the y-axis. A solid teal curve (exposure reduction) and a dashed gray curve (blockable-share ceiling) both rise steeply then level off, climbing from near zero at recurrence 0 to about 80 and 86 percent respectively at recurrence 12, with the reduction curve always sitting just below the ceiling. A horizontal orange dotted line marks the 50 percent H1 threshold, which both curves cross between recurrence 2 and 3.

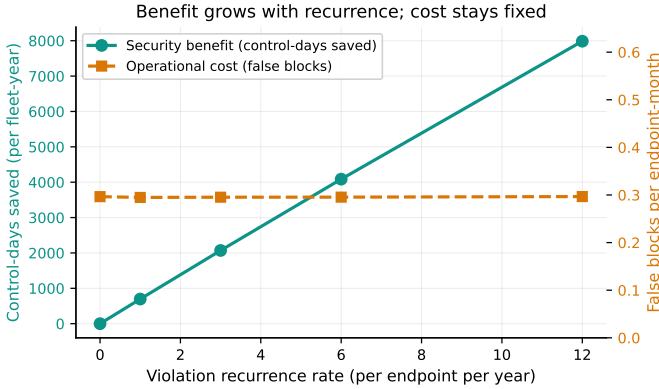


Fig. 2. The security benefit (Criminal Justice Information control-days saved) grows with recurrence while the operational cost (false blocks per endpoint-month) stays flat, so the benefit-to-cost ratio rises across the sweep.

Alt text: Dual-axis line chart with violation recurrence rate per endpoint per year (0 to 12) on the x-axis. The left teal y-axis shows control-days saved per fleet-year (0 to 8000) and its solid teal line rises approximately linearly from 0 at recurrence 0 to about 8000 at recurrence 12. The right orange y-axis shows false blocks per endpoint-month (0 to about 0.6) and its dashed orange line stays flat near 0.30 across the entire sweep. The two lines cross near recurrence 5.

to benefit-dominated as recurrence rises: below roughly one recurrence per year the false-block friction outweighs the exposure saved, while at the reference recurrence the benefit already exceeds the cost more than twofold.

VII. ROBUSTNESS AND SENSITIVITY

The headline numbers are reported at the reference recurrence, but the structural claims are properties of the full sweep, and reading the sweep end to end is what separates a finding that holds across operating regimes from one that holds at a single tuned point. Table III collects the complete per-recurrence response of every reported quantity, so that the

TABLE III
FULL PER-RECURRENCE SENSITIVITY SWEEP (MEAN OVER 25 SEEDS, FROZEN SEEDS 900 TO 924). RECURRENCE IS PER ENDPOINT PER YEAR; REDUCTION R IS THE FRACTIONAL EXPOSURE DECREASE OF PREVENTION OVER DETECTION; EXPOSURE SAVED IS IN CRIMINAL JUSTICE INFORMATION CONTROL-DAYS PER FLEET-YEAR; FALSE BLOCKS ARE PER ENDPOINT-MONTH; BENEFIT-TO-COST IS THE DIMENSIONLESS RATIO OF SAVED EXPOSURE TO TOTAL FALSE-BLOCK COST.

Recurrence r	Reduction R	Exposure saved	False blocks	Benefit/cost
0	0.0000	0.0	0.2965	0.00
1	0.3134	696.5	0.2949	0.39
3	0.5651	2069.9	0.2953	1.15
6	0.7030	4086.2	0.2955	2.27
12	0.8050	7986.6	0.2967	4.42

monotone growth of the reduction and the saved exposure, the flatness of the false-block cost, and the rising benefit-to-cost ratio can be inspected jointly rather than asserted from the reference point alone.

The sweep makes three robustness statements concrete. First, the reduction is monotone in recurrence and bounded above by the blockable share at every row, so the ceiling of Eq. (5) is not an artifact of the reference point: as r rises from zero to twelve the blockable share $r/(r+\rho)$ rises and the realized reduction rises with it but stays beneath it, exactly the recall-scaled relation of Eq. (6). Second, the false-block column is flat across the entire sweep, from 0.2965 at zero recurrence to 0.2967 at twelve, a range that never departs from the constant predicted by Eq. (8), so the recurrence-independence of the cost is not a single-point coincidence but a property of every operating regime. Third, the benefit-to-cost ratio is increasing and crosses unity between one and three recurrences per year, placing the break-even where the closed form of Eq. (9) predicts it; an agency reading the table can locate its own fleet on the recurrence axis and read off whether prevention is benefit-dominated for its churn level rather than relying on the reference figure.

A. Adversarial residual: operating through the unblockable floor

The blockability ceiling is a statement about an honest violation stream, but a guardrail is a control surface and an adversary can choose where to apply pressure. The structural weakness is explicit in Eq. (5): preventive enforcement removes at most the blockable share B , leaving the emergent floor $1 - B = \rho/(r + \rho)$ untouched because there is no change to refuse. An adversary who understands this does not fight the guardrail on its own ground. Rather than attempting a blockable configuration change, which is refused at the door and yields no exposure, the adversary routes the same compliance-relevant effect through an unblockable action: allowing a certificate to lapse, letting a time-based session policy expire, or aging out a credential rather than disabling it by an intercepted edit. Each such action lands in the emergent floor that prevention cannot lower, so an adversary operating entirely within the $1 - B$ residual evades preventive enforcement completely and pays exposure only at the detective cadence, which is exactly the posture prevention was meant to improve upon. The benefit-to-cost analysis is

unchanged for the defender, but the adversarial reading inverts the deployment lesson: the residual that prevention structurally cannot reach is precisely the surface a strategic adversary will prefer, so the emergent floor must be covered by detective controls and proactive lifecycle automation rather than treated as a small leftover. Prevention narrows the attack surface to the unblockable share, and a defense that stops there has merely told the adversary which door is still open; the residual demands a continuous monitor sized to the emergent rate, not the abandonment of detection that an optimistic prevention pitch implies.

VIII. DISCUSSION

The two consequences of preventive policy-as-code point in different directions, and conflating them produces the optimism this study is designed to puncture. Prevention removes the exposure window entirely for the violations it can block, and it does so on every recurrence, so where misconfigurations keep returning it removes a large and growing share of Criminal Justice Information exposure. But prevention cannot reach emergent violations, and an imperfect guardrail lets a residual through, so the reduction is bounded below the blockable share and cannot be pushed to unity by any improvement in the allowlist. A reduction of 0.703 at the reference recurrence is large and worth pursuing, but it is closer to three quarters than to the near-elimination that a naive prevention pitch would suggest, and the right program metric is the blockable-exposure share, not the fraction of changes a guardrail can in principle refuse.

The recurrence result turns this into an actionable ordering. Because the absolute exposure saved grows monotonically with recurrence while the false-block cost stays flat, the benefit-to-cost ratio rises from 0.39 at one recurrence to 4.42 at twelve, so an agency should block its high-recurrence blockable controls first. These are the controls whose misconfigurations keep returning, where detection pays a fresh window each time and prevention pays it none. The same logic gives a guardrail on the other side: where violations are rare and non-recurring, the false-block friction can dominate the exposure saved, so prevention is not uniformly preferable and detection may remain the right posture for low-churn controls.

Three deployment rules follow directly. First, size expectations to the blockable-exposure share, not to unity, when projecting the compliance-exposure benefit of a preventive program; the emergent remainder is an irreducible floor to be managed by other means, such as proactive certificate and credential lifecycle automation rather than admission-time blocking. Second, sequence the rollout by recurrence, because the high-recurrence controls dominate the return and cross the benefit-to-cost break-even first. Third, report security exposure and operational false blocks as the different units they are, rather than netting them into a single score, because they scale differently with recurrence and a netted figure would hide exactly the structure that governs the deployment decision.

IX. THREATS TO VALIDITY

Construct validity. The model abstracts violation arrivals as a Poisson process with fixed detection, remediation, and false-

positive parameters. Real violations may be bursty, correlated across endpoints (a single bad deployment can trip several at once), or clustered around change windows, and real guardrail precision depends on the specific rule set. These would shift absolute exposure and false-block values but not the structural relationships: a blockability ceiling exists whenever any violation is emergent, the absolute benefit grows with recurrence under any arrival process that pays a window per occurrence, and the false-block cost is recurrence-independent whenever benign change volume is set independently of violation recurrence.

The Poisson assumption. Treating recurrences as a homogeneous Poisson process is the cleanest defensible default but is an idealization. If recurrences cluster, the absolute saved exposure at a given mean recurrence would be larger than the model reports, strengthening rather than weakening the recurrence-amplification finding; the monotone direction is robust to the arrival process even where the magnitude is not.

Fixed parameters. The fleet size, horizon, emergent and benign change rates, guardrail recall, and false-positive rate are documented priors, not measurements from any agency, so absolute magnitudes such as 4086.2 control-days saved at the reference recurrence and 0.30 false blocks per endpoint-month should be read as illustrative of the model rather than as field estimates. The comparative findings depend on the qualitative structure of the model rather than on the specific constants.

External validity. Because the priors are not calibrated against a real law-enforcement change record, the absolute values do not transfer directly to a named fleet. The ceiling, the recurrence scaling, and the recurrence-independence of the false-block cost follow from the accounting and are expected to transfer; calibrating the rates against a real CJIS change and audit record is the natural next step and would convert the illustrative magnitudes into estimates.

Statistical validity. Intervals are BCa bootstrap intervals over 25 seeds, appropriate for the small-sample, possibly skewed statistics reported here; we use interval non-overlap rather than significance testing throughout, avoiding the multiple-comparison pitfalls of repeated p -values. Because the study is pre-registered and the evaluation seeds were inspected only once, the reported numbers are not the product of analytic search, and no rate was tuned to clear a threshold.

X. CONCLUSION

Preventive policy-as-code is a strong posture for enforcing security and compliance controls, but its program-level benefit is governed by two facts that a naive prevention pitch hides. A substantial share of violations is emergent and cannot be blocked, setting a hard floor on exposure, and the value of prevention is concentrated where misconfigurations recur, because detection pays a fresh window on every recurrence while prevention pays none. Both facts follow from the exposure accounting and are properties of the policy-as-code approach rather than of any single regime, so they apply equally to NIST 800-53, CIS Benchmark, and CJIS deployments whose violations split into change-driven and emergent parts. In a

pre-registered simulation grounded in the CJIS case study, prevention reduced Criminal Justice Information exposure by 0.703 at the reference recurrence against a binding ceiling of 0.7534, saved exposure that grew monotonically from 696.5 to 7986.6 control-days as recurrence rose from one to twelve per year, and imposed a flat false-block cost of about 0.30 per endpoint-month that the security benefit outgrew, lifting the benefit-to-cost ratio from 0.39 to 4.42. The deployment guidance is correspondingly concrete: block the high-recurrence blockable controls first, size expectations to the blockable-exposure share, and treat the false-block cost as a fixed toll that prevention earns back as churn rises. Calibrating the model against a real CJIS change record is the next step toward turning these structural findings into field estimates.

DATA AVAILABILITY STATEMENT

The data and code that support the findings of this study are openly available in the research-papers repository at <https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper13>. The manuscript sources, figures, analysis code, and frozen result tables are included and regenerate every reported number deterministically from the fixed seeds.

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